

GORYACHINSK, Russian Federation

WMO: 307310

Lat: 52.986N

Lon: 108.291E

Elev: 1612

StdP: 13.86

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.4	-16.2	-25.9	1.4	-19.8	-21.6	1.8	-15.8	18.2	8.8	16.2	8.2	2.1	130	0.650

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	75.4	63.2	71.8	61.8	69.0	60.7	64.7	71.9	63.1	69.7	61.6	67.7	4.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
61.9	87.9	68.5	60.4	83.2	66.8	58.9	78.9	65.2	30.7	72.2	29.5	69.7	28.4	67.9	71.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	13.5	11.8	-23.7	83.4	6.3	4.7	-28.3	86.8	-32.0	89.6	-35.5	92.2	-40.2	95.6		
(5)			-24.0	67.9	6.3	1.6	-28.5	69.1	-32.2	70.0	-35.7	70.9	-40.3	72.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	32.2	1.7	3.3	16.0	31.7	42.4	53.1	60.5	59.5	48.2	35.2	21.2	11.6		
	DBStd	21.71	10.84	10.29	10.71	7.01	5.93	5.43	4.76	4.76	5.96	6.45	9.63	9.06		
	HDD50	7302	1497	1308	1054	548	249	27	0	1	103	459	865	1191		
	HDD65	12001	1962	1728	1519	998	702	359	154	182	503	923	1315	1656		
	CDD50	803	0	0	0	0	12	120	327	294	50	0	0	0		
	CDD65	28	0	0	0	0	0	2	16	10	0	0	0	0		
	CDH74	167	0	0	0	0	1	23	81	61	1	0	0	0		
	CDH80	17	0	0	0	0	0	1	8	8	0	0	0	0		
	WSAvg	4.8	4.7	3.8	3.9	4.2	4.5	3.9	3.9	4.1	5.2	5.4	6.9	6.9		
	PrecAvg	15.7	0.6	0.2	0.2	0.6	1.0	1.7	2.7	2.8	1.7	0.8	1.6	1.6		
(15) Precipitation	PrecMax	21.0	2.2	0.4	1.0	1.3	1.9	4.4	7.1	5.4	3.9	2.4	3.1	4.4		
	PrecMin	10.7	0.2	0.0	0.0	0.0	0.2	0.2	0.6	0.7	0.4	0.2	0.3	0.3		
	PrecStd	3.0	0.5	0.1	0.2	0.3	0.4	0.9	1.7	1.3	0.9	0.5	0.7	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.0	32.2	44.5	55.7	67.1	77.7	80.6	80.1	71.0	55.7	44.9	33.3		
		MCWB	27.5	27.0	35.9	44.2	52.0	59.6	64.2	63.4	57.0	45.9	37.6	28.6		
	2%	DB	25.7	27.4	39.2	49.9	60.9	70.5	75.9	74.5	65.0	51.0	41.1	29.2		
		MCWB	22.8	23.9	32.9	40.5	48.4	58.4	64.1	63.0	56.3	43.0	35.6	25.3		
	5%	DB	20.6	22.9	35.1	45.5	57.1	66.0	72.6	70.9	61.3	47.8	37.7	26.6		
		MCWB	17.9	20.1	30.1	38.2	46.5	56.8	63.0	61.8	53.4	41.7	33.0	23.2		
	10%	DB	16.7	18.6	31.3	42.2	53.7	62.7	69.5	68.3	57.9	45.1	34.7	23.8		
		MCWB	14.6	16.5	27.6	36.0	44.9	54.7	61.7	60.5	51.7	39.6	30.4	20.9		
	0.4%	WB	27.5	27.4	36.5	44.7	53.1	63.1	68.0	66.4	60.4	47.4	38.4	29.2		
		MCDB	29.9	31.6	43.3	55.2	64.1	70.6	76.9	75.1	67.0	54.8	44.1	32.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	22.6	23.7	33.2	41.0	49.9	60.0	65.5	64.3	57.1	44.0	35.9	25.6		
		MCDB	25.5	27.3	38.7	48.7	59.2	67.7	73.2	71.6	63.5	49.3	40.8	28.9		
	5%	WB	17.9	20.2	30.5	38.6	47.7	57.7	63.8	62.7	54.7	42.2	33.3	23.3		
		MCDB	20.3	22.8	34.7	45.0	55.8	64.6	70.3	69.5	60.1	47.4	37.6	26.3		
	10%	WB	14.6	16.5	27.5	36.4	45.5	55.7	62.4	61.4	52.2	39.9	30.6	21.1		
		MCDB	16.5	18.6	31.3	41.8	52.7	61.9	68.5	67.2	57.2	44.8	34.5	23.9		
	5% DB	MDBR	12.4	14.9	15.8	14.9	16.6	15.8	13.9	13.8	13.8	12.5	10.8	9.9		
		MCDBR	12.5	16.6	17.2	19.9	22.7	22.2	19.1	19.5	18.8	17.3	12.8	10.3		
		MCWBR	10.9	13.7	12.3	12.8	13.3	12.1	9.7	10.5	11.0	12.5	10.2	9.0		
		MCWBR	12.2	16.5	16.8	19.1	20.2	18.8	15.6	16.7	16.2	16.7	12.6	10.1		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.8	13.8	12.4	12.8	12.7	11.3	8.9	9.6	10.1	12.4	10.4	9.0		
	Clear-Sky Solar Irradiance	taub	0.262	0.290	0.318	0.365	0.429	0.440	0.428	0.385	0.337	0.308	0.278	0.259		
		taud	2.094	2.079	2.121	2.241	2.121	2.209	2.289	2.393	2.474	2.422	2.158	2.052		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	223	253	272	274	260	259	259	266	268	251	214	196		
		Edh at Noon	23	32	38	39	46	43	39	33	27	23	21	20		
(44) All-Sky Solar Radiation	RadStd		348	672	1135	1546	1807	1915	1759	1480	1087	650	339	237		
			24	34	41	98	110	120	139	70	91	60	29	18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-3.15	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	-3.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page